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TUBE ICE PLANT — 62.4 kWp ON-GRID SOLAR

Single-line diagram · Panel layout · Bill of materials

Project	Solar PV for a tube ice machine (18 kW 3-phase compressor motor + auxiliaries), daytime operation 6–8 h	**Document**	TN-20261019-ICE01 Rev01
System	On-grid, zero export · 96 × LONGi 650 W = 62.4 kWp · Huawei SUN2000-50K-MC0	**Date**	19 October 2026

1. Design basis

Item	Value
Running load (compressor ~17 kW + cutter, pump, condenser fan)	~20–22 kW, 3-phase 400/230 V
Daily energy at 6–8 h operation	~130–175 kWh/day
PV array	62.4 kWp; DC/AC ratio 1.25; ~245 kWh/day average (~89,000 kWh/yr)
Solar-only operation (clear day)	About 09:00–15:30; PV ≥ 22 kW from ~9:00 (~24 kW) to ~15:30
Grid role	Motor start, early/late hours and cloudy periods; no export (smart meter control)
Motor starting	Soft starter / VFD reduces start current from ~190 A (DOL) to ~50–60 A

2. String design

String	Table	Modules	MPPT	Vmp @ 52 °C	Voc @ 0 °C	Isc (STC / +10% rear)
S1	T1	16 × 650 W	MPPT1	≈ 603 V	≈ 835 V	16.5 / 18.1 A
S2, S3	T2, T3	2 × 16 × 650 W	MPPT2	≈ 603 V	≈ 835 V	33.1 / 36.2 A
S4, S5	T4, T5	2 × 16 × 650 W	MPPT3	≈ 603 V	≈ 835 V	33.1 / 36.2 A
S6	T6	16 × 650 W	MPPT4	≈ 603 V	≈ 835 V	16.5 / 18.1 A

Inverter limits: 1,100 V max input; MPPT range 180–1,000 V (rated 600 V); MPPT2/3 accept up to 44 A short-circuit current; 23 A per input. All strings are inside the limits with margin.

3. Single-line diagram

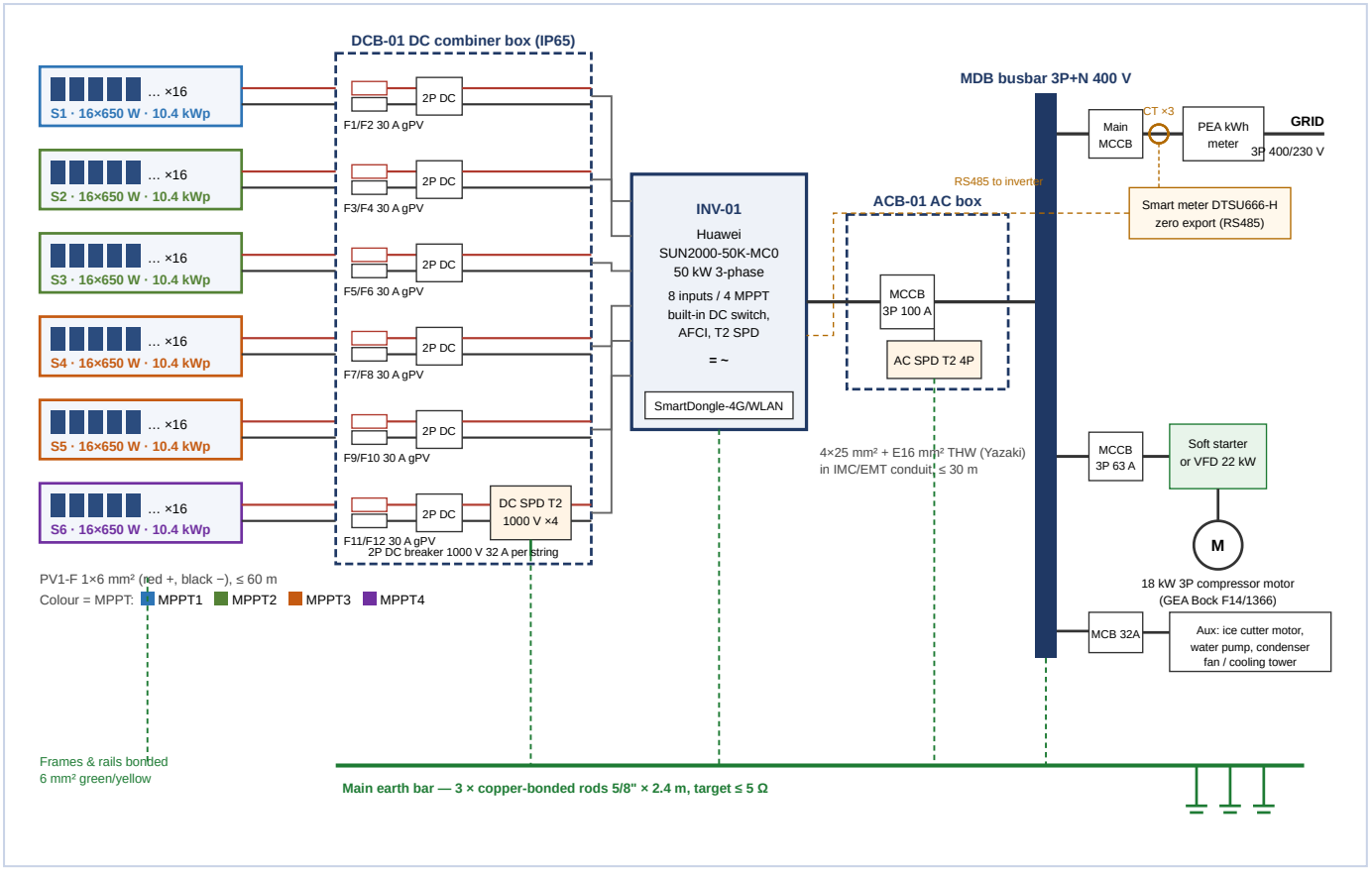


Figure 1 — Single-line diagram. DCB-01 DC combiner box and ACB-01 AC box next to the inverter; the smart meter and CTs at the grid incomer keep export at zero. Earth bonding shown dashed green.

4. Panel layout

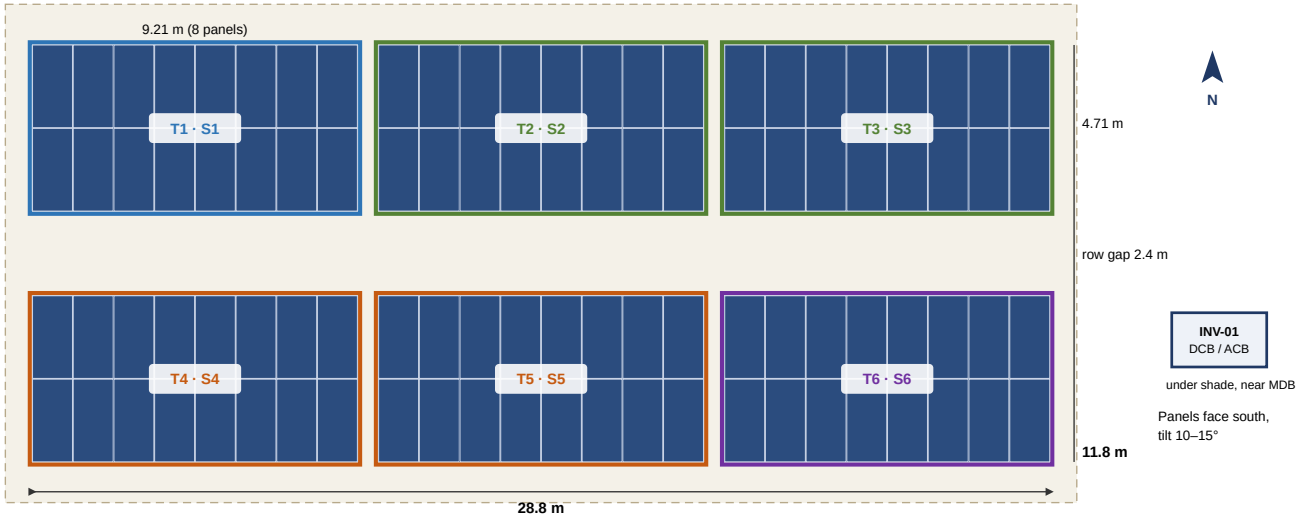


Figure 2 — Ground or flat-area layout: 6 tables (T1–T6) of 2 × 8 panels in portrait, one string per table, colour-coded by MPPT. Array 28.8 × 11.8 m ≈ 340 m² including the 2.4 m row gap and walkways. On a metal-sheet roof the same 96 panels fit about 280 m² of south- or east/west-facing roof, mounted on rails.

5. Bill of materials

No.	Item	Specification	Unit	Qty
A	**PV array and inverter**			
1	PV module	LONGi LR8-66HVD-650M, 650 Wp, anti-glare, back-contact	pcs	96
2	Inverter	Huawei SUN2000-50K-MC0, 50 kW 3-phase, 4 MPPT / 8 inputs, AFCI	set	1
3	Smart meter (zero export)	Huawei DTSU666-H + 3 split-core CTs sized to the main breaker	set	1
4	Monitoring	Huawei SmartDongle-4G (or WLAN-FE), FusionSolar app	set	1
B	**Mounting structure (ground or flat area)**			
5	Mounting tables	Hot-dip galvanized steel / aluminium frame, 6 tables of 2×8 panels portrait, tilt 10–15°	table	6
6	Mid clamps	For 30 mm frame, stainless bolts (incl. 5% spare)	pcs	176
7	End clamps	For 30 mm frame, stainless bolts (incl. 5% spare)	pcs	50
8	Footings	Helical pile Ø89 mm or concrete pad per STR-02 design, 8 posts per table	post	48
C	**DC side**			
9	PV cable, red	PV1-F 1×6 mm², 1.8 kV DC, TÜV, LINK	m	300
10	PV cable, black	PV1-F 1×6 mm², 1.8 kV DC, TÜV, LINK	m	300
11	PV connectors	Same make as the module connectors (pairs, incl. spares)	pair	18
12	DC combiner box	KJL steel enclosure IP65 with gland plate, DCB-01	set	1
13	gPV fuse + holder	SUNTREE 30 A gPV 1000 V DC, 10×38, both poles of each string	pcs	12
14	DC breaker	SUNTREE 2P 32 A 1000 V DC, one per string	pcs	6
15	DC surge protection	SUNTREE Type II 1000 V DC, one per MPPT	pcs	4
D	**AC side**			
16	AC box	KJL steel enclosure IP55, ACB-01	set	1
17	Inverter output breaker	CHINT MCCB 3P 100 A, 25 kA	pcs	1
18	AC surge protection	SUNTREE Type II 4P 40 kA 385 V AC	pcs	1
19	AC cable (phases + neutral)	Yazaki THW 25 mm² (4 cores × 30 m)	m	120
20	AC earth cable	Yazaki THW 16 mm² green/yellow	m	30
21	Conduit and fittings	IMC/EMT 1¼", boxes, straps, glands	lot	1
E	**Motor and load side (recommended)**			
22	Soft starter or VFD	22 kW 3-phase for the 18 kW compressor motor, with bypass contactor	set	1
23	Ice machine feeder breaker	CHINT MCCB 3P 63 A in the MDB	pcs	1
F	**Earthing**			
24	Earth rods	Copper-bonded 5/8" × 2.4 m, with clamps and inspection pits	pcs	3
25	Earth conductor	16 mm² green/yellow to main earth bar	m	50
26	Bonding conductor	6 mm² green/yellow for frames, rails and boxes	m	100
G	**Services**			
27	Installation	Mounting, electrical installation, labelling	lot	1

28	Testing and commissioning	IEC 62446 tests: polarity, Voc, Isc, insulation, earth resistance; zero-export check	lot	1
29	PEA documents	Notification and documents for the zero-export system	lot	1

6. Notes and items to confirm

- Loads are estimated from the nameplates in the photos (GEA Bock F14/1366 compressor driven by an 18 kW motor; Müller gear motor nameplate not readable). Confirm the auxiliary motor ratings and whether the condenser is air-cooled or uses a cooling tower.
- Grid connection assumed 3-phase 400/230 V with enough capacity to start the motor without solar. Confirm the main breaker rating so the CTs and cables can be sized exactly.
- Cable lengths assume the inverter within 30 m of the MDB and the array within 60 m of the inverter; quantities are adjusted after the site survey.
- Mounting assumes a ground or flat area of about 30 × 12 m facing south. For a roof installation, rails and clamps replace items 5 and 8 after a roof survey.
- Off-grid / backup option (if the grid is weak or absent): Solis S6-EH3P50K-H hybrid inverter with 60 kWh FlexCore-ID battery, with the same PV array.